

Joshua Cox

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Education

University of Illinois Urbana-Champaign

Master of Science, Mechanical Engineering

GPA: 3.83 / 4.00

Expected: December 2025

Bachelor of Science, Mechanical Engineering | Minor, Computer Science

GPA: 3.61 / 4.00

December 2022

Skills

Programming Languages: Python, C/C++, MATLAB, Java, C#, Microsoft VBA

Software: Simulink, PyTorch, Docker, Git, ROS, Godot, pygame, Unity CARLA Simulation, Gazebo Simulation, LaTeX

Hardware: STM32 Nucleo boards, TI C2000 MCU boards, Stretch3, UR3 Robot arm, CRS Robot arm, Arduino, Raspberry Pi

Research Experience

Lunar Autonomy Challenge | Human-Centered Autonomy Lab

November 2024 - Present

- Founded team to compete in NASA software challenge to autonomously navigate and map lunar surface using a rover
- Implemented navigation strategy employing both a high and low-level planner, creating local object avoidance methods, querying rover's battery usage, and determining heuristics on when to recalculate path
- Created 2D exploration simulation from scratch with python and pygame in order to test team's navigation approach

Handle Haptics | Human-Centered Autonomy Lab

February 2024 - Present

- Designed, created CAD for, and 3D-printed parts for mounting strategies to attach vibration sensors onto a Hello Robot Stretch3 robot to better communicate directions and signals to guided users
- Rapidly prototyped initial designs including a technique to attach Nintendo Joy-Cons in various positions
- Created final mount design where user can feel vibrational motors through flexible TPU-printed slots

Anomaly Detector Informed Learning | Human-Centered Autonomy Lab

August 2023 - Present

- Composed simulation environment utilizing CARLA to demonstrate data gathering and learning of mobile robots
- Tested if using an anomaly detector improves imitation learning performance on mobile robots by identifying unfamiliar scenarios and prompting an expert to intervene

Work Experience

Embedded Systems Engineer Intern | Engineering System Design Lab | Urbana, IL

May 2024 – August 2024

- Completed hardware and software tasks to develop a new testing setup for a hinge-integrated multifunctional structures for attitude control (MSAC) scheme for satellites utilizing piezoelectric strain actuators
- Programmed software in C++ for STM32 microcontroller Nucleo-H755ZI-Q for the purpose of enabling advanced PWM signals, logging SD card data, and communicating with multiple accelerometers
- Tested types of signal filters to produce sin signals from period-changing PWM signals as inputs for the actuators
- Created various MATLAB scripts that analyzed signal frequency response and ensured proper signal generation

Systems Engineer Intern | Quartus Engineering | San Diego, CA

January 2022 – May 2022

- Tested flamethrower robot to find its most efficient parameters by discussing an in-house experimental setup with customer to ensure monetary, timeline, and tangible goals were met
- Built and calibrated camera assemblies in a cleanroom by preparing future materials, adhering to a strict construction schedule, troubleshooting software issues as needed, and observing customer requests
- Costed and designed a customer's heavy lifter assembly and corresponding lift test tooling. Documented, assembled, organized, and cleanroom-tested the physical end product using an overhead crane

Project Experience

Spaceshot Rocketry Team

August 2021 – December 2022

- Assembled and tested active roll control systems composed of rack-and-pinion servo motors, ensuring adequate clearances required during flight operation
- Advanced on-board embedded system that saves sensor data and adjusts servos in response to active control algorithm utilizing C++ and pair-programming
- Validated new avionics stack performance onboard test rocket, which successfully guided rocket to 10,000 feet, deployed control flaps when necessary, and recorded flight telemetry for future analysis

NASA Micro-g Neutral Buoyancy Experiment Design Team

August 2019 – June 2020

- Designed astronaut tool tolerant of operating with regolith dust that collects lunar rock samples
- Built and tested low-fidelity prototype to assess possible assembly schemes
- Composed and arranged sections of technical papers describing our team's tool and task plan
- Established test protocols according to NASA standards for validating tool effectiveness in the Neutral Buoyancy Lab